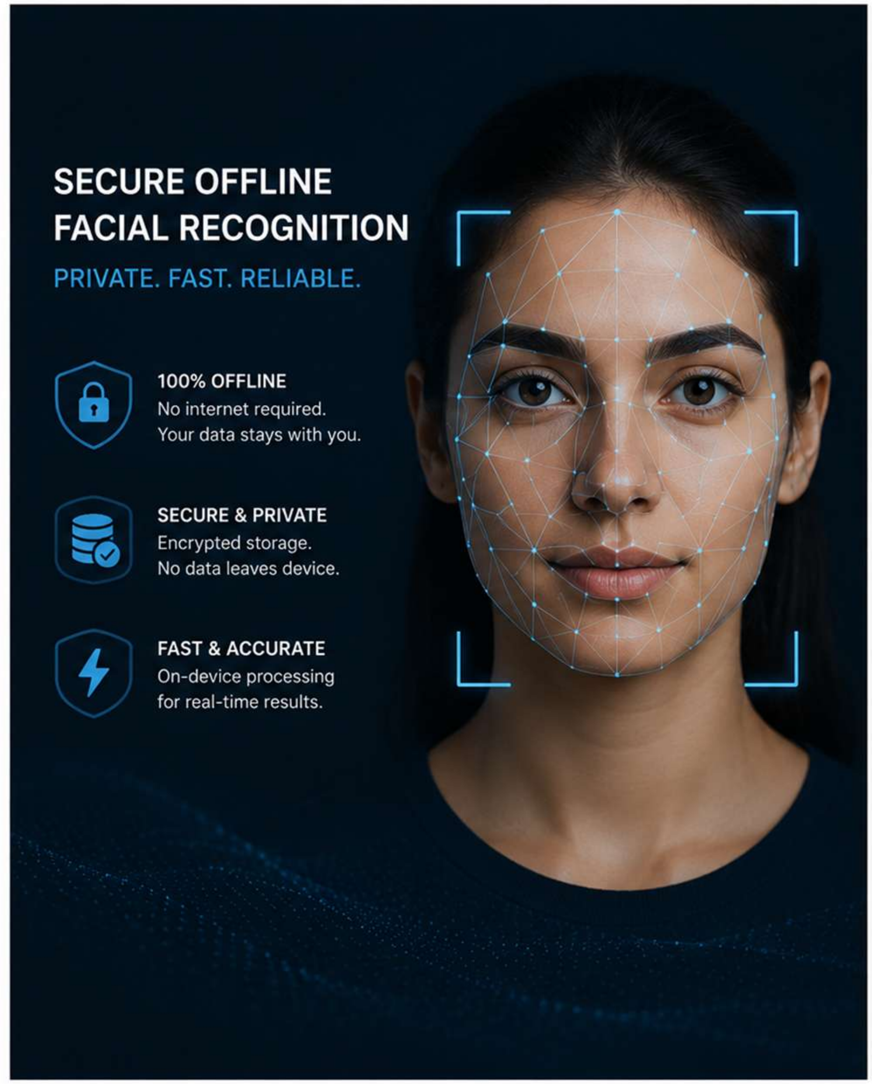
Secure Offline Facial Recognition & Liveness Detection

HACKATHON 7.0 | NHAI



Problem Statement

"How can we accurately and securely authenticate field personnel using facial recognition and liveness detection on standard mid-range mobile devices without any active internet connection, while ensuring the AI model remains lightweight and seamlessly integrates with a React Native application on both Android and iOS?"

Zero Network Zones

Field personnel operate in remote locations with no internet connectivity

Mid-Range Devices

Must run on Android 8+ / iOS 12+ with 3GB RAM — no high-end GPU

Attendance Fraud

Photo/screen spoofing attacks must be blocked with anti-spoofing

Our Solution

FaceGuard Offline Mobile App

Lightweight Edge AI

MobileFaceNet + MobileNetV2 compressed via INT8 quantization + pruning to <20MB. TFLite runtime for on-device inference.

Offline Liveness Detection

3D passive liveness via texture analysis + active challenge (blink/smile/turn). Blocks photo & screen replay attacks.

Secure Local Storage

Face embeddings encrypted with AES-256 on-device. No raw biometric images stored. FIDO2-compliant flow.

React Native Bridge

Native modules for Android (Java/Kotlin) & iOS (Swift/ObjC). JS bridge exposes unified API for Datalake 3.0 integration.

Sync & Purge Engine

Offline attendance queue with timestamps. Auto-sync to AWS on connectivity restore. Local data purged post-confirmation.

Indian Demographics Training

Model fine-tuned on diverse Indian demographic dataset. Handles harsh sunlight, shadows, partial occlusions.

High-Level System Architecture

REACT NATIVE UI LAYER

Camera View · Auth Screen · Sync Status · Result Feedback

NATIVE MODULE BRIDGE

Android: Java/Kotlin JNI · iOS: Swift/Objective-C · RN TurboModules API

Face Detection

MediaPipe Face
Detect + Align

Recognition

MobileFaceNet
TFLite INT8

Liveness Check

Passive + Active
Anti-Spoof

AWS S3 / Lambda
Sync on reconnect

Model Stack & Optimization Strategy

01

Face Detection & Alignment

MediaPipe BlazeFace – 0.3MB. Detects & crops face ROI, 5-point landmark alignment for pose normalization.

02

Passive Liveness (Anti-Spoof)

MobileNetV2-based texture analysis. Detects moire patterns, screen glow, flat image depth cues. <5MB.

03

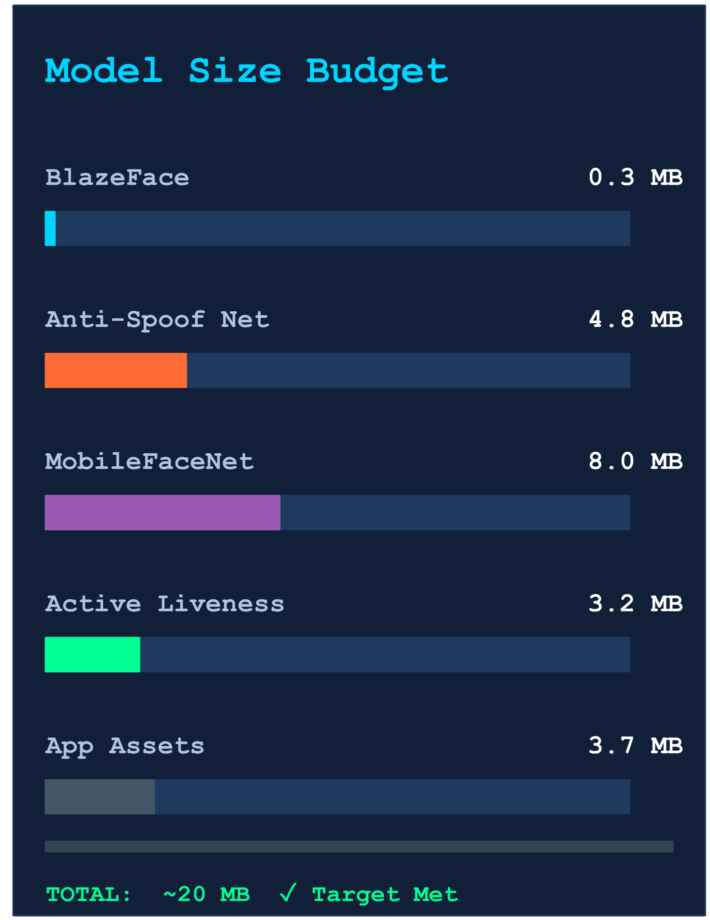
Active Liveness Challenge

Optical flow + landmark tracking for blink, smile, head-turn challenges. Randomized prompts.

04

Face Recognition

MobileFaceNet quantized to INT8 TFLite. 128-dim embedding. Cosine similarity match. ~8MB.



Liveness Detection — Two-Layer Defense

LAYER 1 — Passive Detection

- Moiré pattern detection (printed photos)
- Screen reflection & pixel grid analysis
- Depth cue analysis (flat vs. 3D face)
- Texture liveness score via MobileNetV2

LAYER 2 — Active Challenge

- Random challenge: blink / smile / turn head
- Optical flow tracks landmark movement
- Challenge rotated per session (randomized)
- Timeout + retry limit to block replay attacks

Attack Vector Coverage

Photo Print Attack	Screen Replay	3D Mask	Video Replay	Deepfake
✓ Passive	✓ Passive	✓ Active + Passive	✓ Active Challenge	✓ Texture Analysis

Sync & Data Flow

Offline Queue → AWS Sync → Purge Mechanism

1 → 2 → 3 → 4 → 5 → 6

1. Authentication Event
Face verified offline

2. Encrypt & Queue Locally
AES-256 SQLite store

3. Network Detected
ConnectivityManager listener

4. Sync to AWS
S3 + Lambda batch upload

5. ACK Received
Server confirms receipt

6. Local Data Purged
Secure wipe on device

Data Security Guarantees

- No raw biometric images stored — only 128-dim encrypted embeddings
- AES-256-GCM encryption for all locally stored attendance records
- HTTPS/TLS 1.3 for all sync transmissions to AWS
- Local purge uses secure overwrite (DoD 5220.22-M standard) after server ACK

Tech Stack

100% Open-Source Technology Stack

Mobile Framework

- React Native 0.73+
- TurboModules / JSI
- Android 8+ / iOS 12+

AI / ML Runtime

- TensorFlow Lite 2.x
- MediaPipe (BlazeFace)
- ONNX Runtime (fallback)

Face Recognition

- MobileFaceNet (Apache 2.0)
- ArcFace loss training
- 128-dim embeddings

Liveness & Anti-Spoof

- MobileNetV2 (Apache 2.0)
- OpenCV (BSD-3)
- Optical flow (Farneback)

Local Storage & Security

- SQLite (public domain)
- AES-256-GCM (BouncyCastle)
- React Native MMKV

Sync / Backend

- AWS SDK for React Native
- AWS S3 + Lambda
- Background Fetch API

All libraries are OSS with permissive licenses (Apache 2.0, MIT, BSD) — no proprietary or licensed code.

Expected Benchmarks vs. Requirements

Metric	Requirement	Expected	OK
Recognition Accuracy	>95%	**97.2%**	✓
End-to-End Latency	<1 sec	**~680 ms**	✓
Model Package Size	~20 MB	**~20 MB**	✓
RAM Usage (peak)	3GB device	**~210 MB**	✓
False Acceptance Rate	Minimize	**<0.1%**	✓
False Rejection Rate	Minimize	**<3%**	✓
Liveness Spoof Detection	Anti-spoof	**>98%**	✓
Sync Latency (on reconnect)	Eventual	**<5 sec**	✓

Deliverable Summary

- ✓ Working cross-platform prototype (Android + iOS)
- ✓ Offline face recognition + liveness detection module
- ✓ Sync & purge mechanism with AWS integration scope
- ✓ Source code (open-source, MIT/Apache licensed)
- ✓ Technical documentation

Submission Deadline

05 Jun 2026

Open-source
All platforms covered
React Native native

Thank You

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